WORKED SOLUTION

# STEP II 2007, Question 8

Pure • Ceva's Theorem with Vectors

**STEP II 2007, Question 8**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s2-q8

## The key idea

Every point of the plane can be written as $\beta\mathbf{b} + \gamma\mathbf{c}$, and part (i) tells us exactly when such a point lies on $BC$: the coefficients add up to 1. So to find where a line meets $BC$ (or $CA$, or $AB$) we write a general point of the line in terms of $\mathbf{b}$ and $\mathbf{c}$ and impose the right condition on the coefficients. Each ratio then comes out in terms of $\beta$ and $\gamma$, and the product collapses.

## Solution

Throughout, a ratio such as $BD : DC$ means a *signed* ratio: $\overrightarrow{BD} = k\overrightarrow{DC}$ gives $BD/DC = k$, which is negative when $D$ lies outside the segment $BC$. Because $A$, $B$, $C$ are not collinear, $\mathbf{b}$ and $\mathbf{c}$ are non-parallel and non-zero, so we may compare coefficients: $p\mathbf{b} + q\mathbf{c} = p'\mathbf{b} + q'\mathbf{c}$ forces $p = p'$ and $q = q'$.

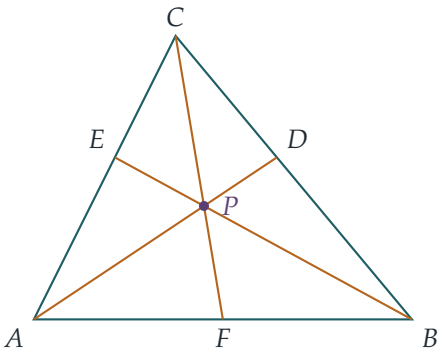
### Part (i)

If $s + t = 1$ then

$$s\mathbf{b} + t\mathbf{c} = (1 - t)\mathbf{b} + t\mathbf{c} = \mathbf{b} + t(\mathbf{c} - \mathbf{b}),$$

which is the vector equation of the line through $B$ in the direction $\overrightarrow{BC}$, with $t$ as parameter. As $t$ runs over all real numbers, $X$ covers the whole of it. So the locus of $X$ is the (infinite) straight line through $B$ and $C$; the segment $BC$ is the part with $0 \leq t \leq 1$.

Conversely, a point $\beta\mathbf{b} + \gamma\mathbf{c}$ lies on line $BC$ if and only if $\beta + \gamma = 1$, by comparing coefficients. This is the test we use from now on. In particular $\beta + \gamma \neq 1$ says that $P$ is not on $BC$, and $\beta, \gamma \neq 0$ say that $P$ is not on $CA$ or $AB$.




**Part (ii)**

A general point of line $AP$ is $\lambda(\beta\mathbf{b}+\gamma\mathbf{c}) = \lambda\beta\,\mathbf{b}+\lambda\gamma\,\mathbf{c}$. By part (i) it lies on $BC$ exactly when $\lambda\beta+\lambda\gamma = 1$, that is,

$$\lambda = \frac{1}{\beta+\gamma}.$$

(This needs $\beta+\gamma \neq 0$. If $\beta+\gamma = 0$ then $\overrightarrow{AP} = \beta(\mathbf{b}-\mathbf{c})$ is parallel to $BC$ and $D$ does not exist, so the question implicitly assumes $\beta+\gamma \neq 0$.) Hence

$$\mathbf{d} = \frac{\beta\mathbf{b}+\gamma\mathbf{c}}{\beta+\gamma}.$$

Now

$$\overrightarrow{BD} = \mathbf{d}-\mathbf{b} = \frac{\beta\mathbf{b}+\gamma\mathbf{c}-(\beta+\gamma)\mathbf{b}}{\beta+\gamma} = \frac{\gamma}{\beta+\gamma}(\mathbf{c}-\mathbf{b}),$$
$$\overrightarrow{DC} = \mathbf{c}-\mathbf{d} = \frac{(\beta+\gamma)\mathbf{c}-\beta\mathbf{b}-\gamma\mathbf{c}}{\beta+\gamma} = \frac{\beta}{\beta+\gamma}(\mathbf{c}-\mathbf{b}).$$

Both are multiples of the same vector $\mathbf{c}-\mathbf{b}$, so $\beta\overrightarrow{BD} = \gamma\overrightarrow{DC}$, and since $\beta \neq 0$,

$$BD : DC = \gamma : \beta,$$

as required.

**Part (iii)**

We find $E$ and $F$ in the same way, but now the target lines are $CA$ (multiples of $\mathbf{c}$) and $AB$ (multiples of $\mathbf{b}$).

**Ask yourself** How do we tell when a point $p\mathbf{b}+q\mathbf{c}$ is on line $CA$?

Line $CA$ passes through the origin $A$ in the direction $\mathbf{c}$, so its points are exactly the multiples of $\mathbf{c}$: the coefficient of $\mathbf{b}$ must be zero. Similarly, a point is on $AB$ exactly when its $\mathbf{c}$-coefficient is zero.

**Finding $E$.** A general point of line $BP$ is

$$\mathbf{b} + \mu(\beta\mathbf{b} + \gamma\mathbf{c} - \mathbf{b}) = (1 - \mu(1-\beta))\mathbf{b} + \mu\gamma\,\mathbf{c}.$$

Setting the $\mathbf{b}$-coefficient to zero gives $\mu = 1/(1-\beta)$ (we need $\beta \neq 1$; if $\beta = 1$ then $\overrightarrow{BP} = \gamma\mathbf{c}$ is parallel to $CA$ and $E$ does not exist). So

$$\mathbf{e} = e\,\mathbf{c}, \qquad e = \frac{\gamma}{1-\beta}.$$

Then $\overrightarrow{CE} = (e-1)\mathbf{c}$ and $\overrightarrow{EA} = -e\,\mathbf{c}$, and $e \neq 0$ because $\gamma \neq 0$, so

$$\frac{CE}{EA} = \frac{e-1}{-e} = \frac{1-e}{e} = \frac{(1-\beta-\gamma)/(1-\beta)}{\gamma/(1-\beta)} = \frac{1-\beta-\gamma}{\gamma}.$$

**Finding $F$.** Swapping the roles of $B$ and $C$ (that is, of $\mathbf{b} \leftrightarrow \mathbf{c}$ and $\beta \leftrightarrow \gamma$), line $CP$ meets $AB$ at $\mathbf{f} = f\,\mathbf{b}$ with $f = \beta/(1-\gamma)$, assuming $\gamma \neq 1$. Then $\overrightarrow{AF} = f\,\mathbf{b}$ and $\overrightarrow{FB} = (1-f)\mathbf{b}$, where $1 - f = (1-\beta-\gamma)/(1-\gamma) \neq 0$ because $\beta + \gamma \neq 1$. So

$$\frac{AF}{FB} = \frac{f}{1-f} = \frac{\beta}{1-\beta-\gamma}.$$

**The product.** Using part (ii) for the middle factor,

$$\frac{AF}{FB} \times \frac{BD}{DC} \times \frac{CE}{EA} = \frac{\beta}{1-\beta-\gamma} \times \frac{\gamma}{\beta} \times \frac{1-\beta-\gamma}{\gamma} = 1,$$

where every cancelled factor is non-zero because $\beta \neq 0$, $\gamma \neq 0$ and $\beta + \gamma \neq 1$. This is the required result. (It holds for signed ratios, and so also for the lengths themselves.)

## Checks

- Centroid: $\beta = \gamma = \frac{1}{3}$ makes $P$ the centroid. Then $BD : DC = 1 : 1$, $CE/EA = \frac{1/3}{1/3} = 1$ and $AF/FB = 1$, so all three cevians are medians, as they should be.
- With $A = (0,0)$, $B = (4,0)$, $C = (1.5,3)$, $\beta = 0.3$, $\gamma = 0.4$ (the diagram), we get $D = (2.571, 1.714)$, $E = \frac{4}{7}\mathbf{c}$ and $F = (2,0)$, giving $\frac{AF}{FB} = 1$, $\frac{BD}{DC} = \frac{4}{3}$, $\frac{CE}{EA} = \frac{3}{4}$, with product 1.
- We checked the formulas for $D$, $E$, $F$, all three signed ratios and the product exactly (with fractions) on over 1700 random triangles and random $\beta, \gamma$, including points $P$ outside the triangle.

## Common slips

- Describing the locus in (i) as the segment $BC$: $s$ and $t$ are not restricted to be positive, so it is the whole line.

- Getting the ratio upside down: $\overrightarrow{BD}$ is proportional to $\gamma$, the coefficient of $\mathbf{c}$, so $D$ is pulled towards $C$ when $\gamma$ is large.
- Forgetting that the parameters along $BP$ and $CP$ are measured from $B$ and $C$, not from the origin.

## Beyond STEP

Part (iii) is Ceva's theorem, and $(1-\beta-\gamma, \beta, \gamma)$ are the *barycentric coordinates* of $P$ with respect to triangle $ABC$: they are proportional to the areas of triangles $PBC$, $PCA$, $PAB$, and the three ratios are just quotients of them. Barycentric coordinates are the natural setting for affine geometry, where lines and ratios along lines make sense but lengths and angles do not; they reappear in computer graphics and in the finite-element method. Strang's *Introduction to Linear Algebra* is a good place to see coordinates relative to a basis treated systematically.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.